

coating, and white light may be used as an aid for examining the complete coverage of the coating. If the bubble is acceptable, the part in question may be released, but if the bubble turns out to be a defect, the part has to be rejected and the defect recorded.

After the safe launch of the application process, its stability regarding bubble formation has to be assessed. While the visual inspection can be performed without destroying the assembly, this does not apply without restrictions when the physical and/or chemical properties or resistances are determined. Depending on the layer thickness, a coating film achieves its maximum property values after a more or less long drying time. The property values have to be tested, in accordance with DIN 46449, at the earliest 96 hours after becoming tack-free. Due to its importance, the following is cited from DIN 46449; section 5.2 'Time of the tests':

"The tests according to sections 5.5 to 5.12 are carried out, unless otherwise mentioned for the individual test procedures, on unaged test specimen, in case of air-drying coatings between 48 and 96 hours after reaching drying stage 3 (according to DIN 46456 Sheet 1, Edition January 1970, Section 6.7; VDE 0360 Part 1165) and in case of oven-drying coatings at the earliest 24 hours after drying."

Drying/curing of a minimum of 96 hours at ambient temperature (after becoming tack-free) is recommended in order to ensure the drying and curing even of slightly thicker layers, e.g., at/under components or drip noses.

In case of oven curing, a minimum of 24 hours as mentioned above have to be observed. Depending on the component geometry, layer thickness, etc more than 24 hours may be necessary under certain circumstances until the maximum property values are achieved.

Visual inspection

After the coating and drying/curing is completed, the coating result is inspected visually for its evenness and

TABLE I Test Parameters (Examples) Applied After Coating	
Contour stability	Respecting of specified coating limits
Wetting	Coating applied on all surfaces as required with a particular eye on critical component leads
Coating defects	Signs of orange peel, film delamination or wetting defects (island formation)
Layer thickness and distribution of layer thickness	Unusual accumulations of coating (coating runs, coating streaks) or extremely thin coating films
Coating runs	Particularly with dip coating, coating runs may form in the emersion area which, according to the installation requirements of the assembly, may be acceptable (if necessary, contact the designer)

completeness. This inspection can be made based on random samples or by verifying individual pieces if the contours or demands are of a complex nature. For large quantities, or in case of in-line productions, optical control systems are employed to a growing extent. These optical control systems use a scanning technique similar to that of automatic component mounting. The scope of the visual inspection results from the specifications (customer requirements) and from internal requirements such as the coating plan and the quality parameters. The visual inspection may be done without referring to any optical aids—thus by the naked eye—or by means of a microscope or magnifying lens. For comparative tests, J-STD-001D recommends a 1.75- to 4-fold magnification.

If the conformal coating contains a UV marker or a fluorescent dyestuff, it is recommended to carry out the inspection under so-called 'black light.' Should a microscope be used for optical inspections, a ring light has proved to be a good choice, since its segments can be individually connected or disconnected. It allows, for example, to selectively fade out debilitating reflections. These ring lights are also available with the appropriate UV-LEDs.

'Black light' is a colloquial term for a UV-A radiation generated by low-pressure gas discharge lamps or LEDs in the wavelength range between 350 nm and 370 nm. This UV-A radiation excites fluorescent substances, e.g., some papers with added pigments or dyestuffs, to glow. This 'black light' inspection, however, is neither suited to detect microscopic pinholes or bubbles

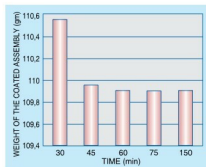


Fig. 2: Example of determination of the dryness stage through weight loss (drying of an assembly at 80°C)

nor to determine the layer thickness.

It should be noted here that not just the coated areas have to be checked but also such areas which must remain uncoated; any splashes in this area, for example, are easily detected by means of black light.

Adhesion test

The adhesion of the coating film after drying/curing can be tested by using various methods. An overview of common adhesion tests is given in Table II.

Determination of drying stage

The determination of the solvent content by checking the weight serves the purpose of easily ascertaining the drying stage after the evaporation of solvents, yet it does not contribute to ascertaining the degree of curing in terms of cross-linking of polymers. Within this test, the coated assembly is weighed after a defined drying time; e.g., after 30 minutes at 80°C. Following this, the respective weight is determined in intervals of 15-minute drying time (the drying temperature